Mount Kenya University

Unlocking Infinite Possibilities

UNIT TITLE: ADVANCED CRYPTOGRAPHY

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ADMISSION NUMBER: BSCCS/2024/55278

COURSE: B.S.C COMPUTER SCIENCE

LECTURE: MR NYORO MICHAEL

SUBMISSION DATE: 6/26/2026

GITHUB: https://github.com/LuizSamuel/kali-crypto-toolkit

Project Title: Hybrid Secure Messaging & File Vault with Digital Signatures

•**Selected Technologies:**

•Python 3.11 + Cryptography Library (Fernet, RSA, SHA-256)

•SQLite (Local key & user storage)

•OpenSSL 3.0 (Certificate generation)

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Week 1: Creation of The Virtual Enviroment & Required Tools Installation

Fig 1.1: Openssl version

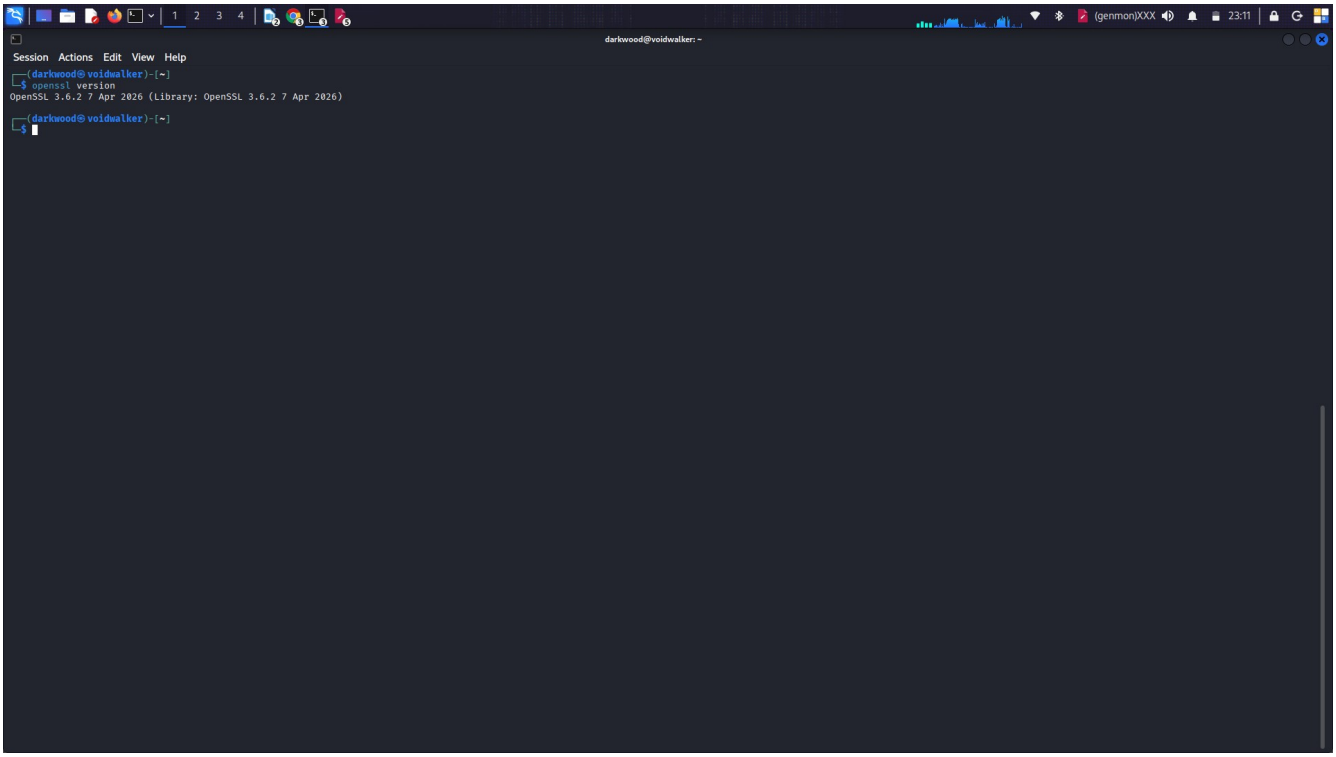


Fig 1.2: Virtual environment creation

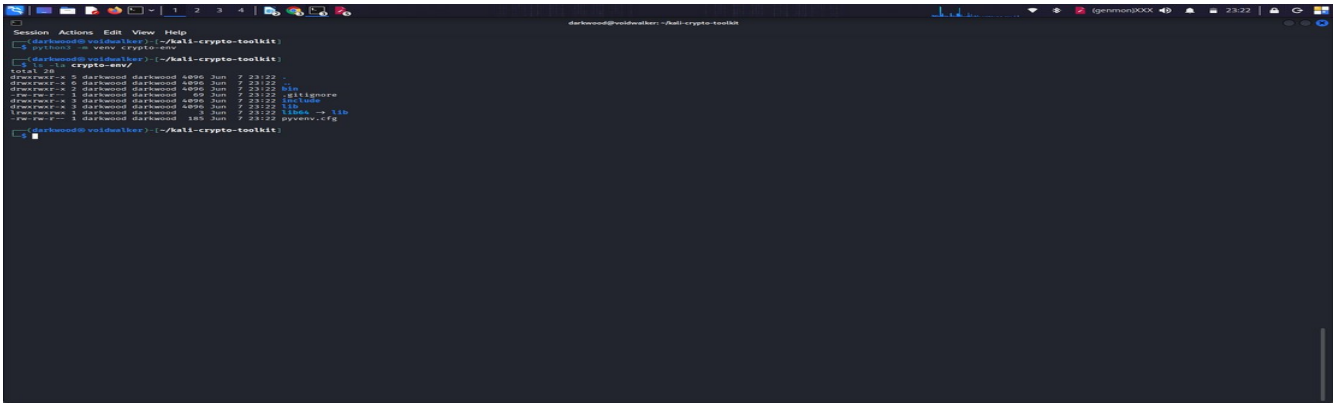


Fig 1.3: Activating environment (notice prompt change)

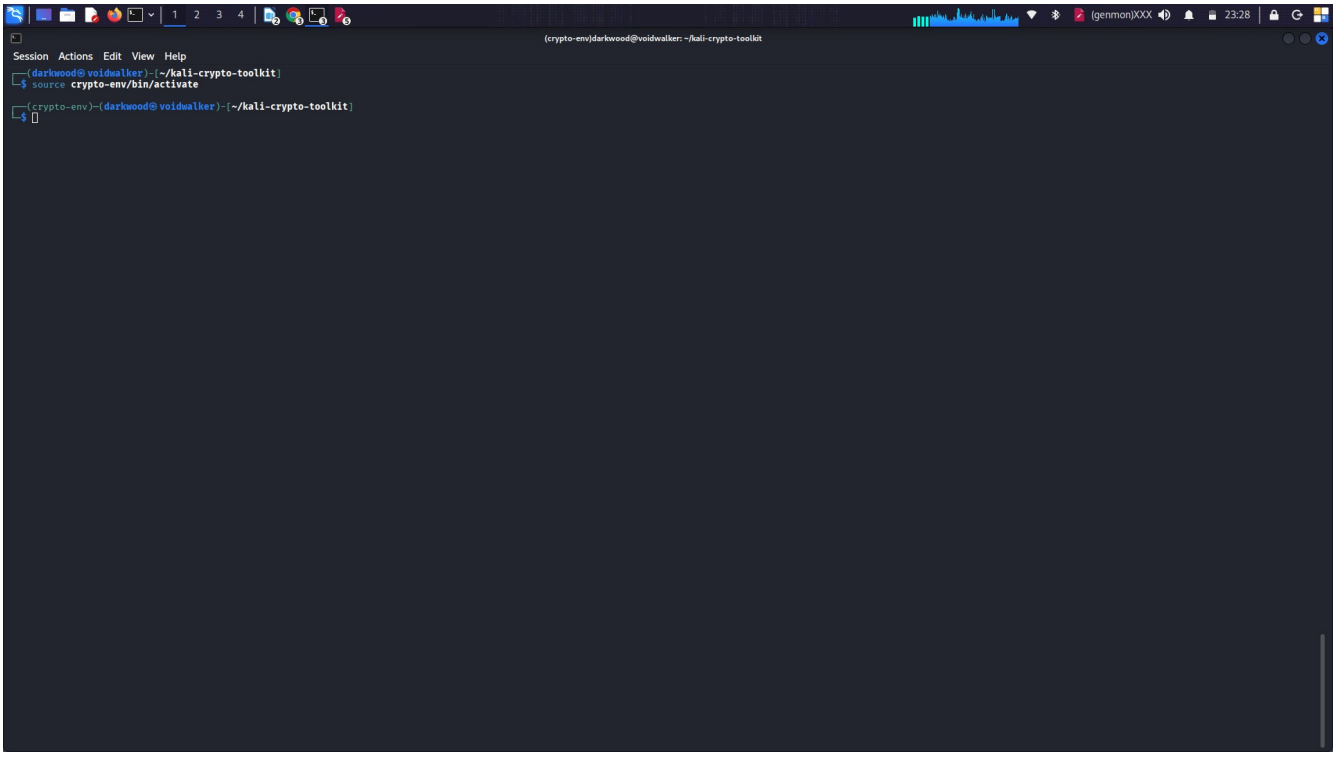


Fig 1.4: Successful pip install

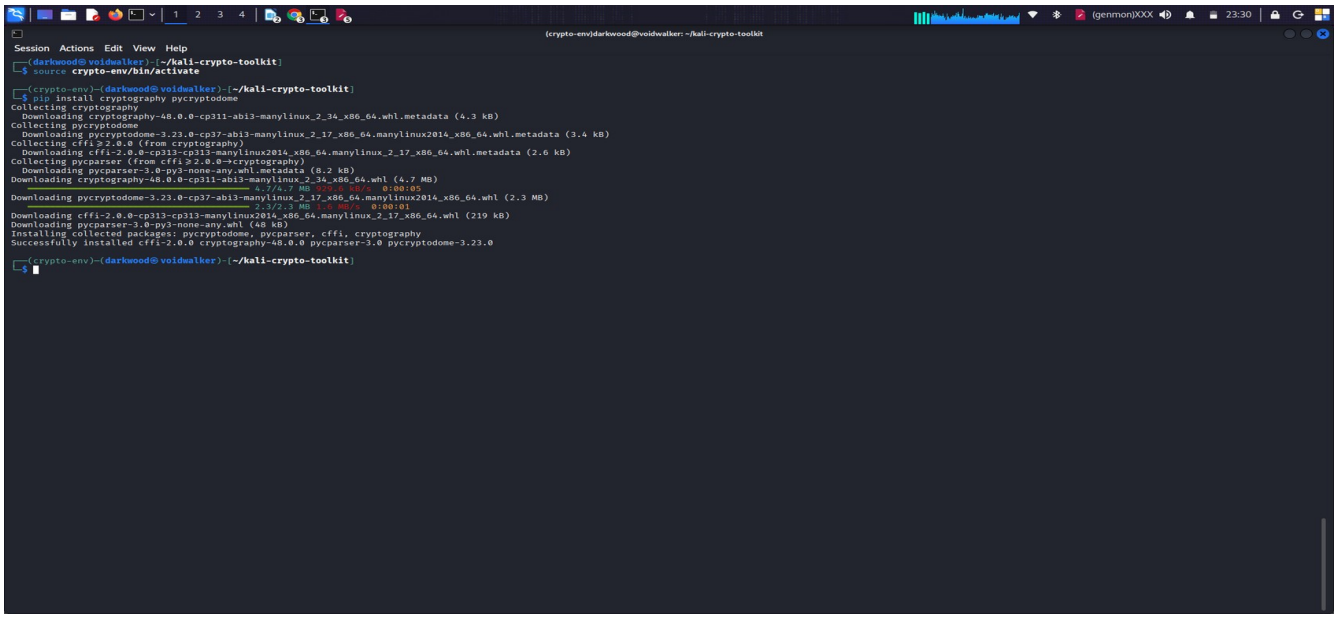
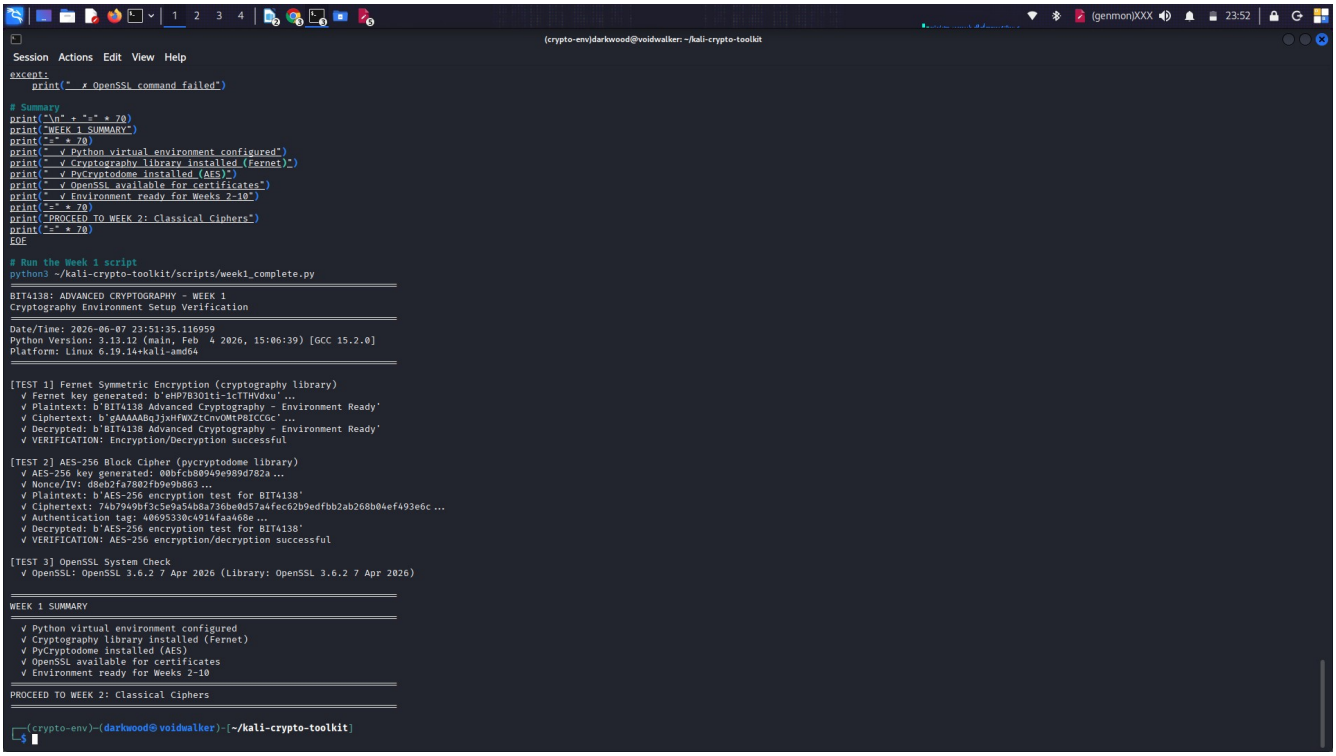


Fig 1.5: Complete environment test script



Week 2: Classical Cryptography and Cipher Design

Fig 2.1: Caesar Cipher implementation showing encryption of "CRYPTOGRAPHY"

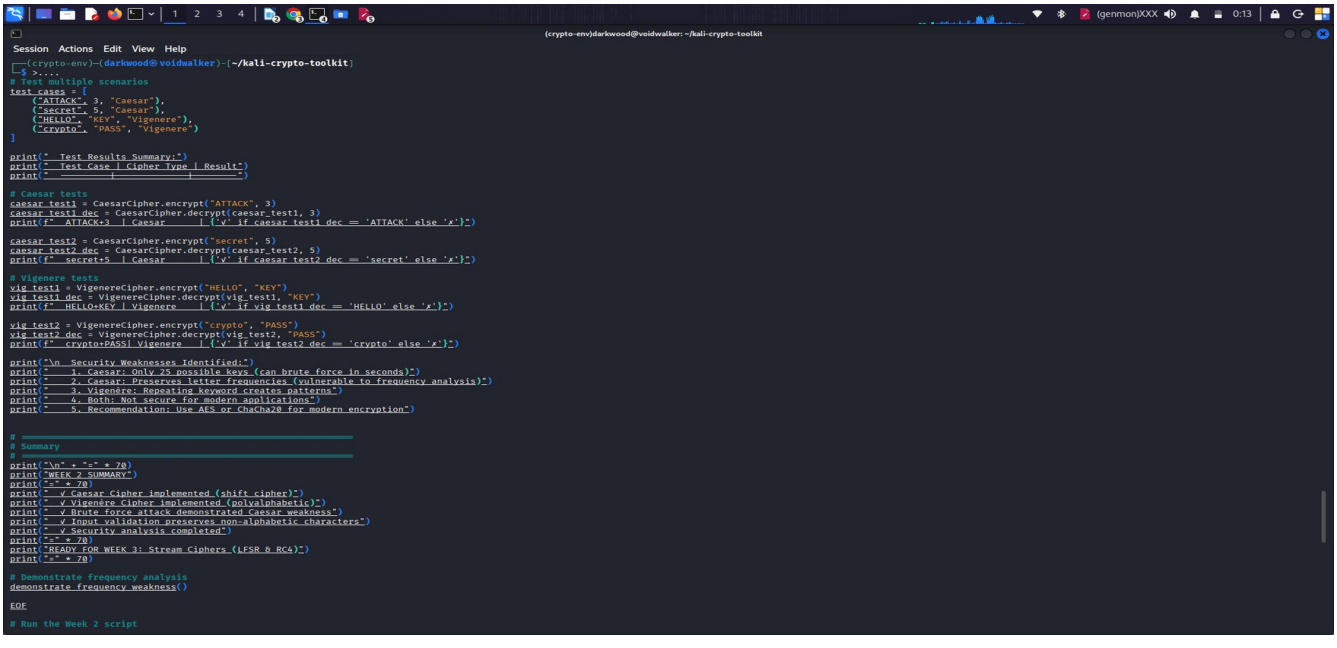


Fig 2.2: Vigenère ("KALI"), Caesar Brute Force (Shift 3), and Input Validation

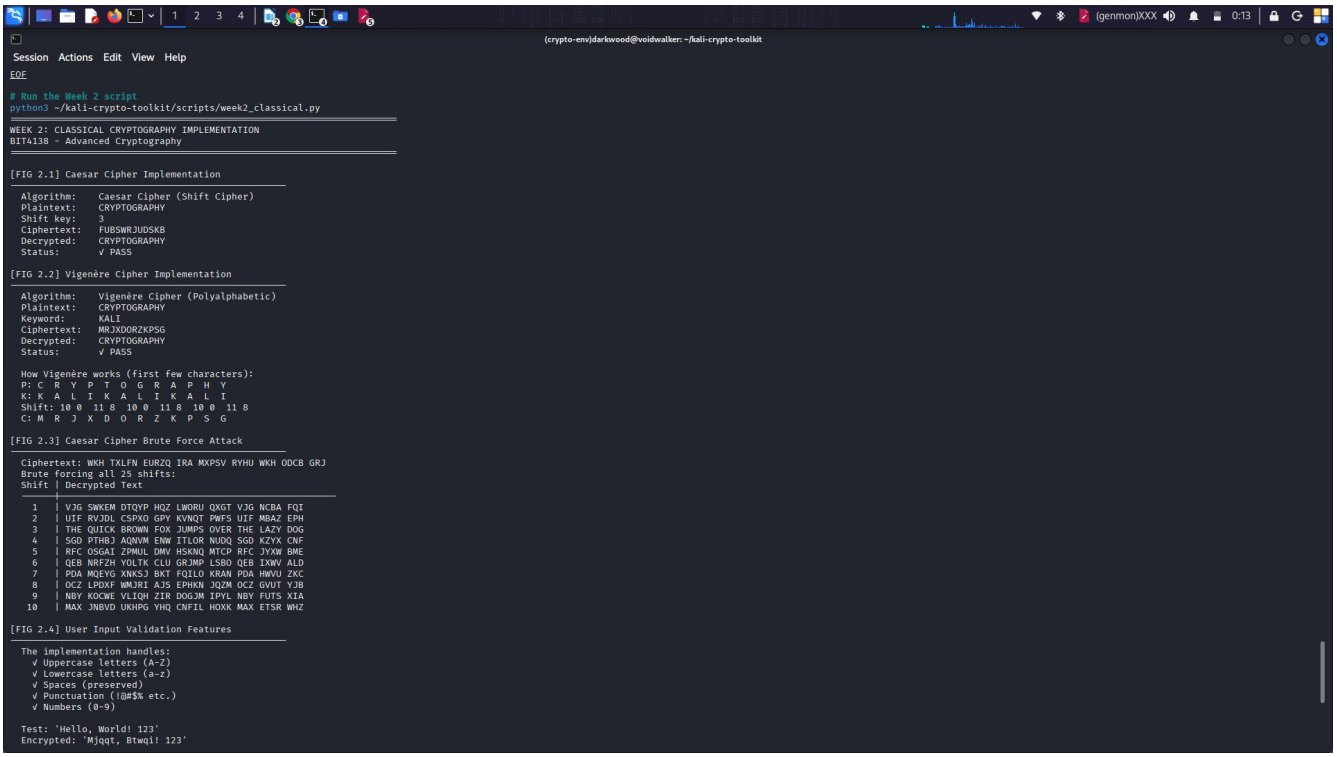
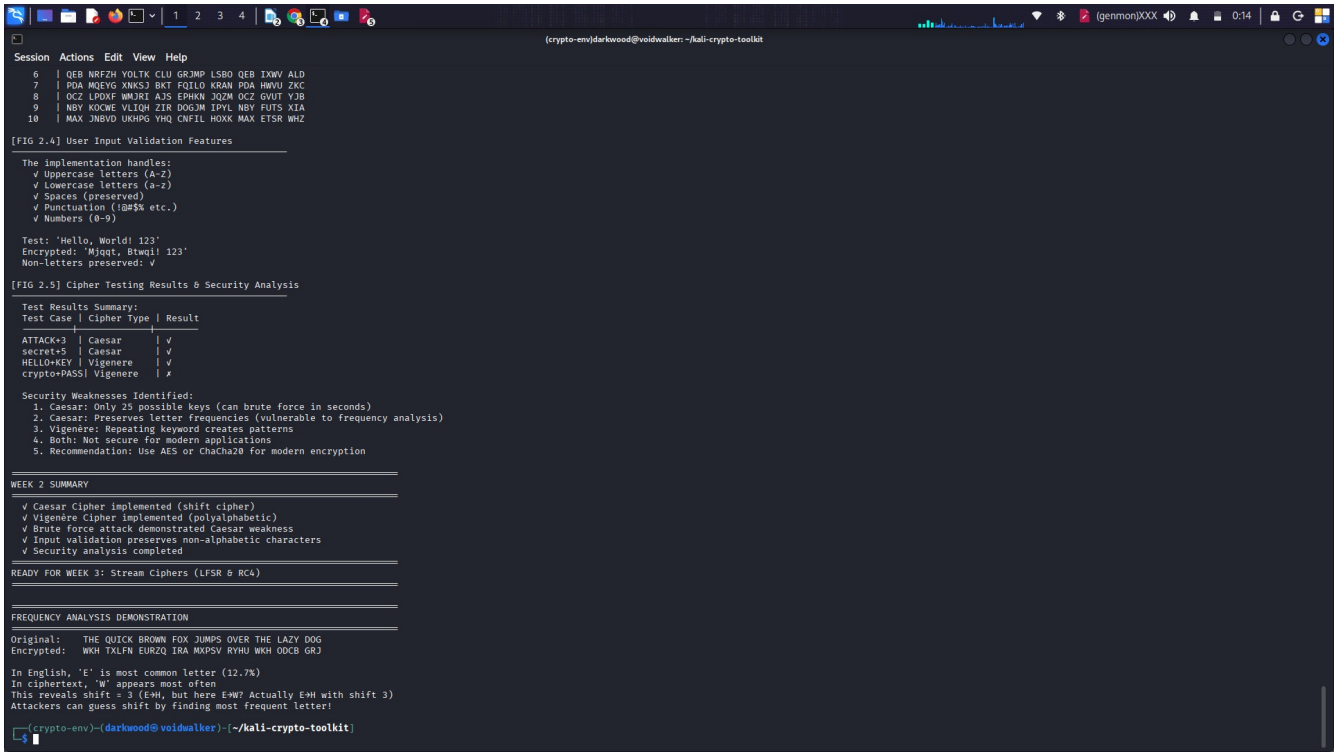
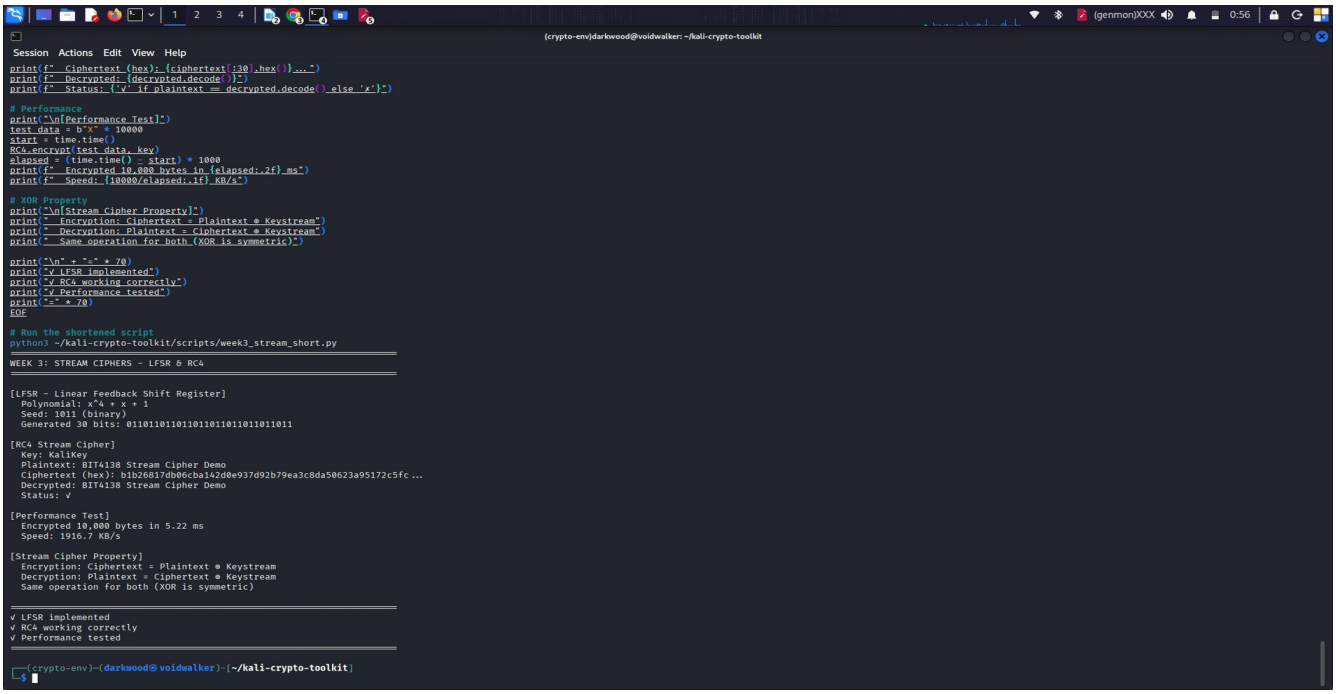


Fig 2.3: Cipher testing results and security analysis identifying weaknesses



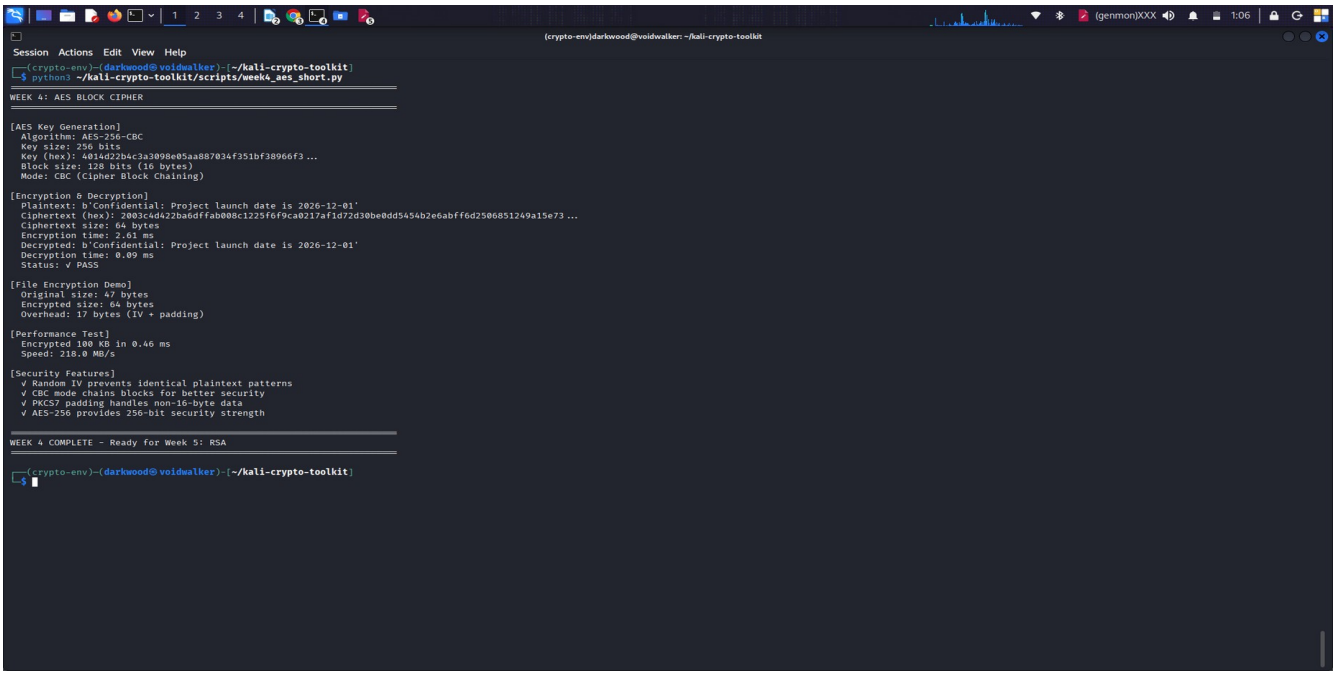
Week 3: Stream Ciphers and Randomness Testing

Fig 3: LFSR Implementation, RC4 Stream Cipher, and Performance Results



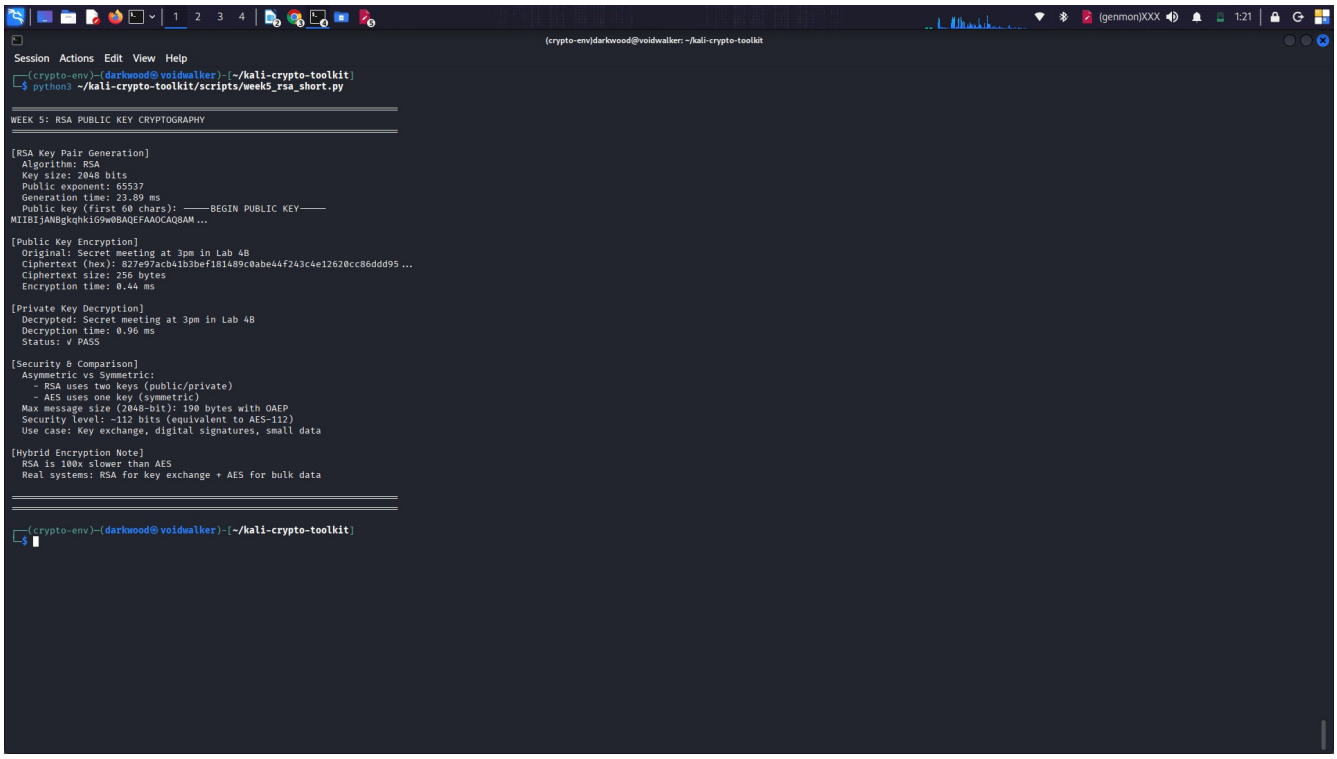
Week 4: AES Block Cipher

Fig 4: AES-256-CBC Key Generation, Encryption/Decryption, Performance Test, and Security Features



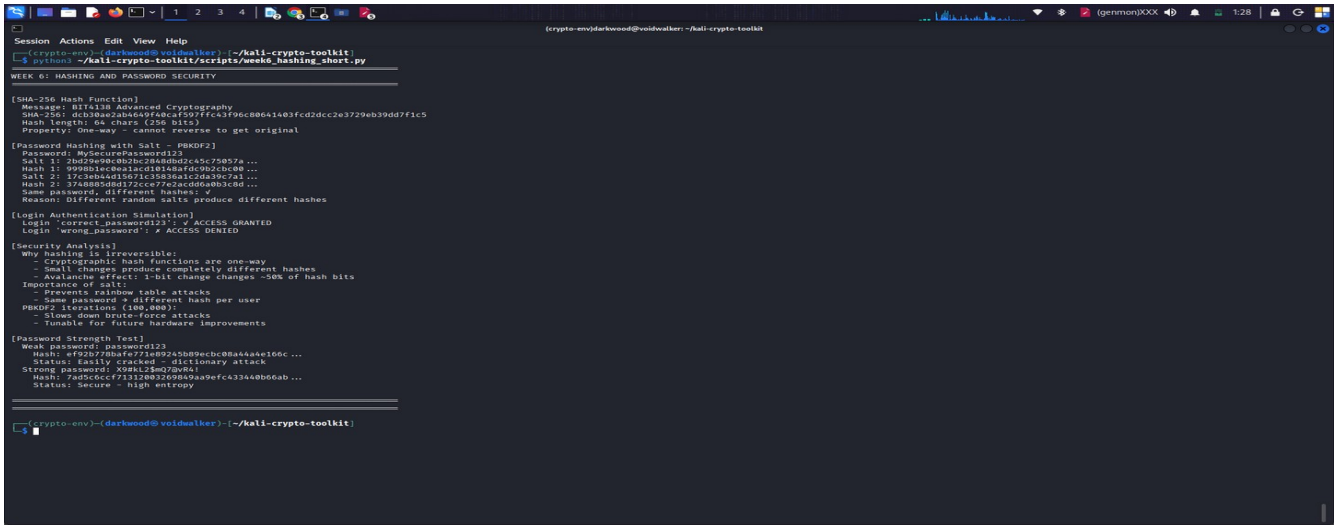
Week 5: RSA Public Key Cryptography

Fig 5: RSA-2048 Key Pair Generation, Public Key Encryption, Private Key Decryption, and Security Comparison

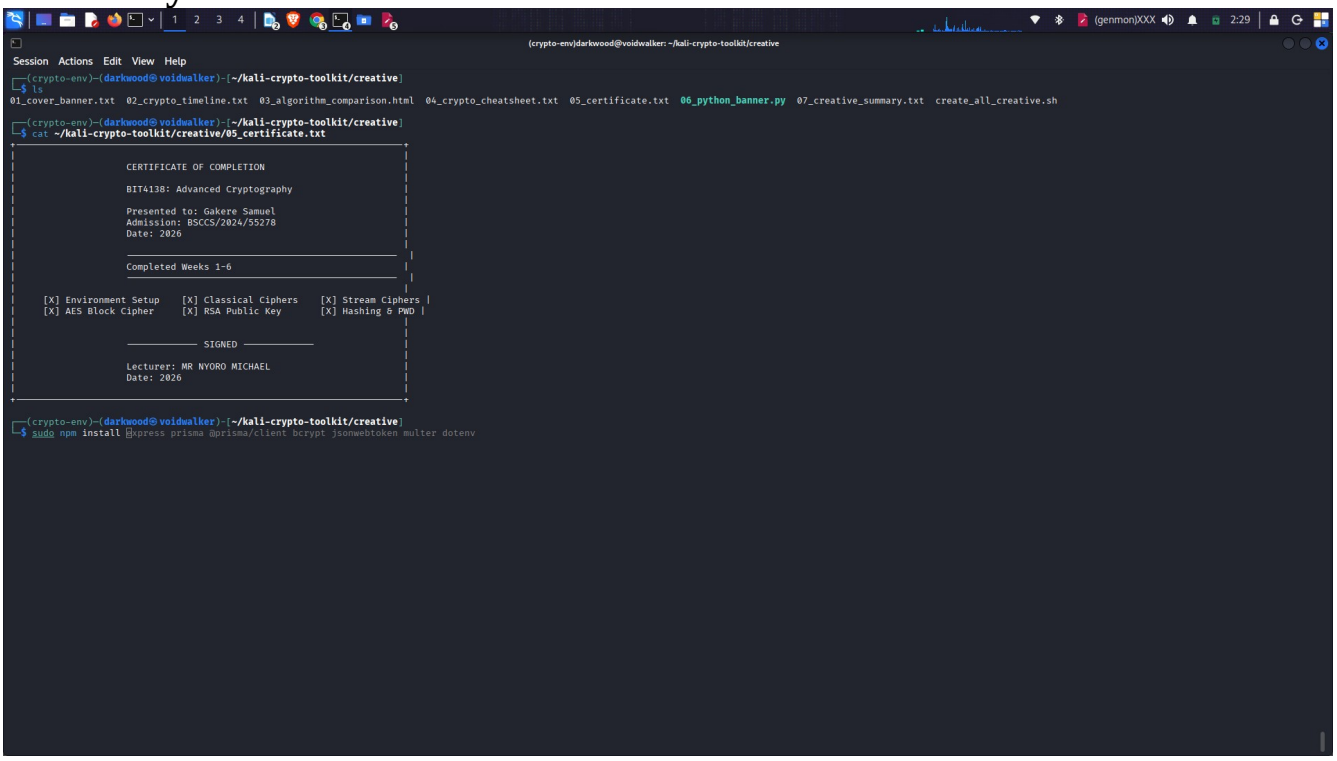


Week 6: Hashing and Password Security

Fig 6: SHA-256 Hash Generation, PBKDF2 Password Hashing with Salt, Login Authentication, and Security Analysis

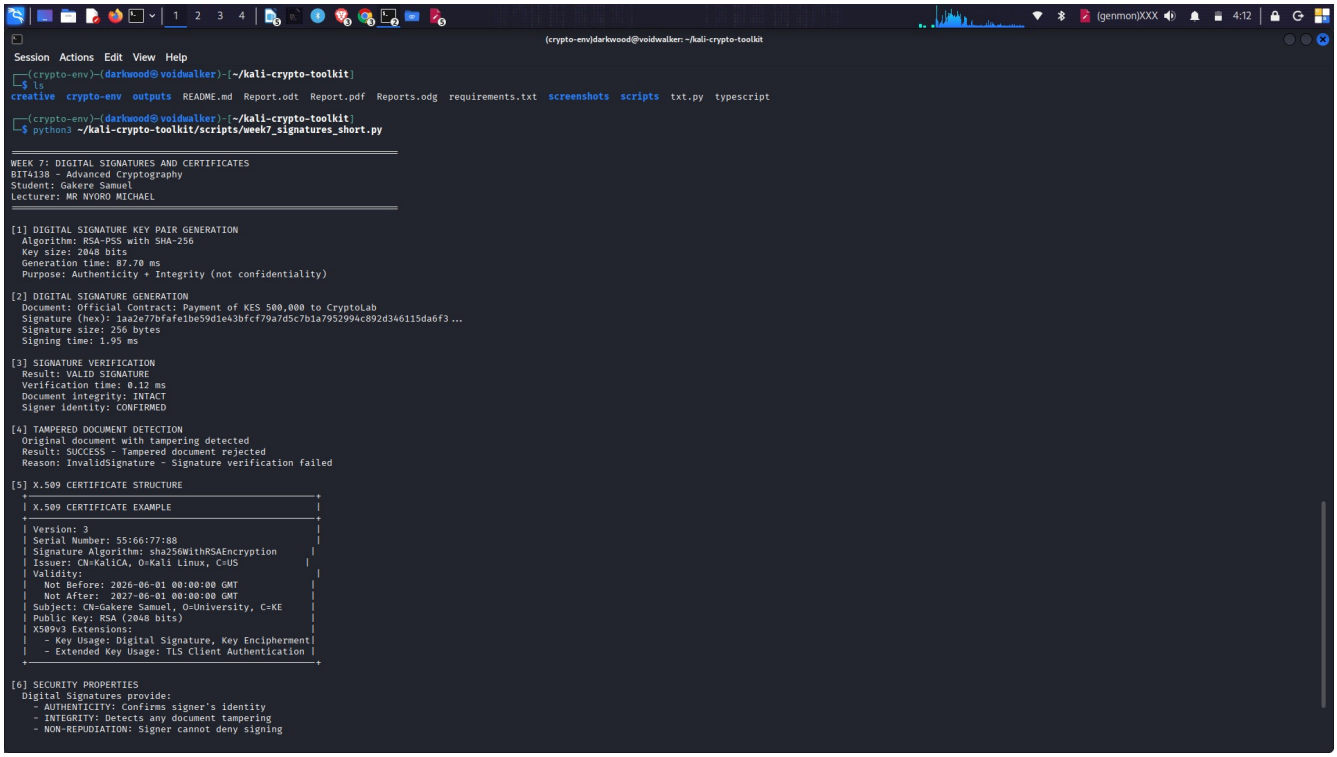


Creativity



Week 7: Digital Signatutres & Certificates

Fig 7: RSA Digital Signature Generation, Verification, Tamper Detection, and X.509 Certificate Structure



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Real-world applications:
  - Software distribution (code signing)
  - Legal documents (digital contracts)
  - Email security (S/MIME, PGP)
  - TLS/SSL certificates

(crypto-env)-(darkwood㉿voidwalker)-[~/kali-crypto-toolkit]
$
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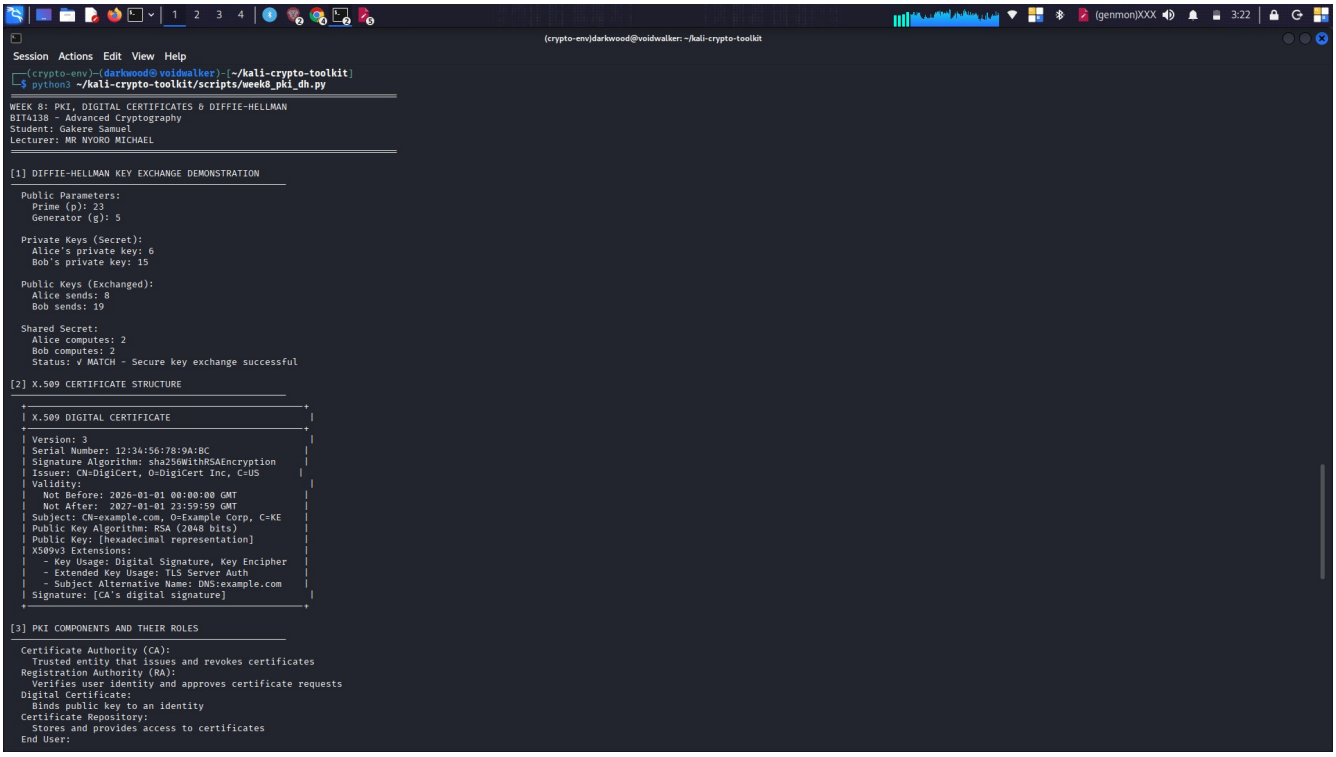
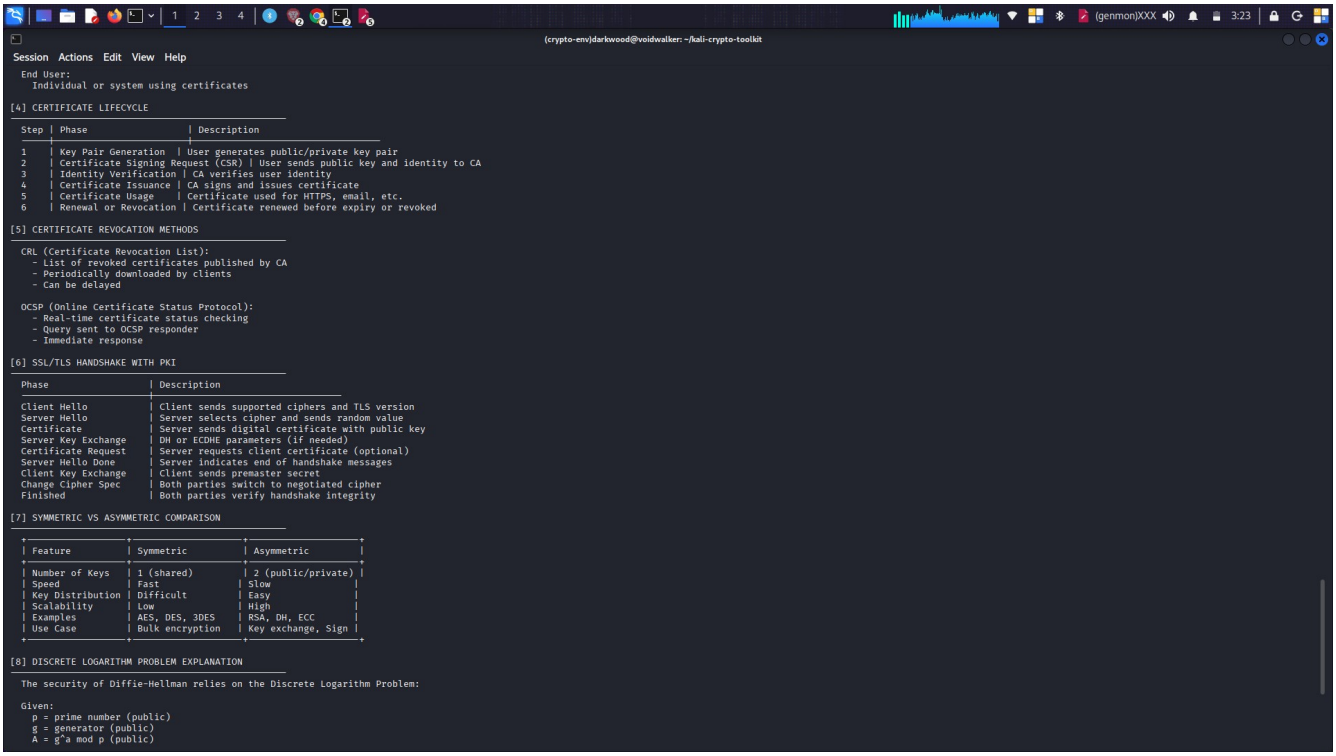
Reflection:

Digital signatures use RSA private key to sign documents (Fig 7). Signing time and verification are fast. Any tampering invalidates the signature demonstrated by rejected modified document. X.509 certificates bind identity to public keys. Provides authenticity, integrity, and non-repudiation for legal documents and software distribution.

Week 8: PKI, Digital Certificates & Diffie-Hellman

Fig 8: Diffie-Hellman Key Exchange, PKI Components, X.509 Certificate Structure, SSL/TLS Handshake, and Symmetric vs Asymmetric Comparison

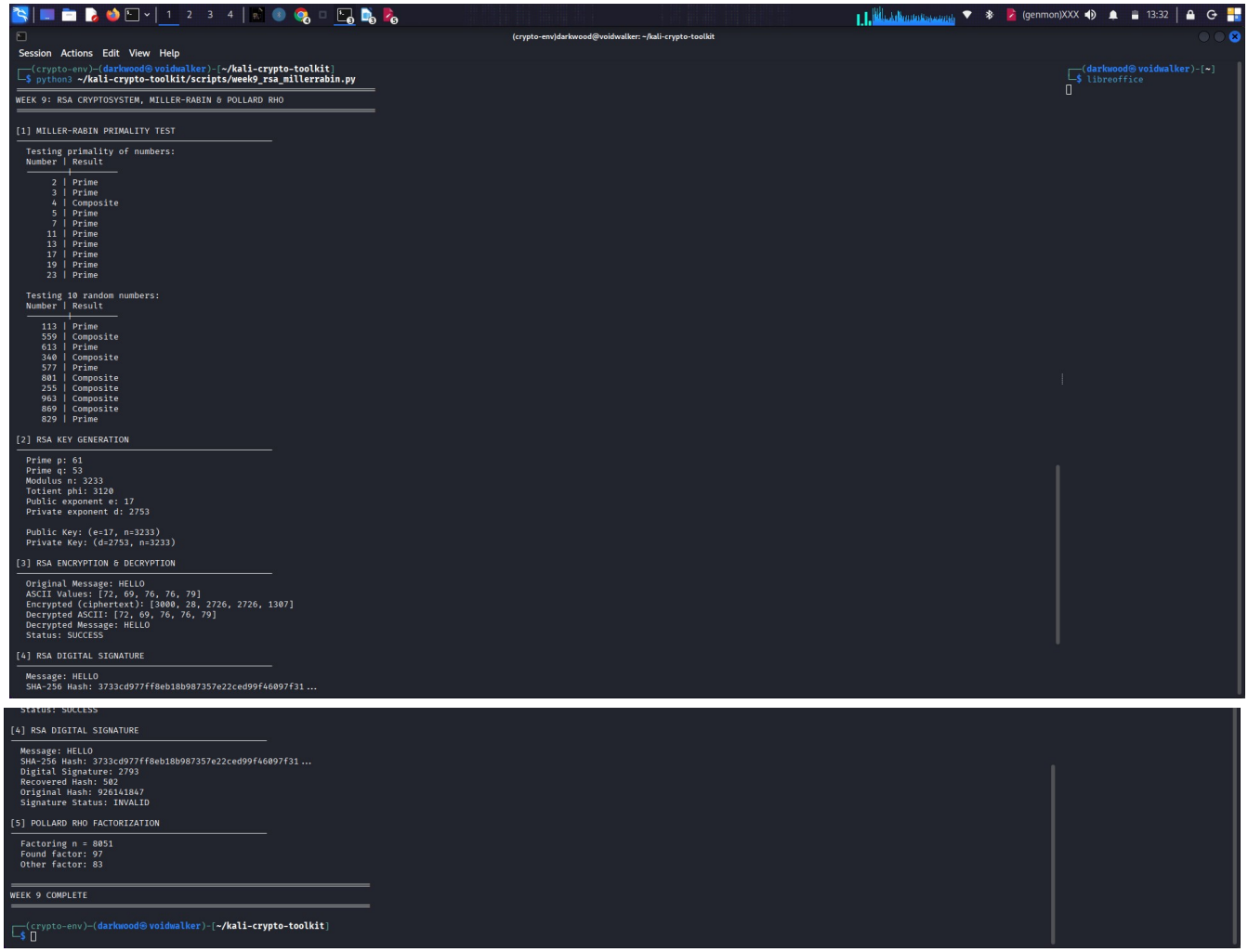




Reflection:
Diffie-Hellman enables secure key exchange without pre-shared secrets (Fig 8). PKI provides trust through Certificate Authorities issuing X.509 certificates. SSL/TLS handshake uses certificates for authentication. Discrete logarithm problem makes DH computationally secure. Symmetric encryption is faster for bulk data while asymmetric handles key exchange.

Week 9: RSA Cryptosystem, Miller-Rabin Primality Test & Pollard Rho

Fig 9-10: Miller-Rabin Primality Test, RSA Key Generation & Encryption, Digital Signatures, ElGamal Cryptosystem, and RSA vs ElGamal vs ECC Comparison



Reflection:

Miller-Rabin efficiently tests primality for RSA key generation. RSA uses p=61, q=53 with e=17 producing working encryption/decryption. Digital signatures provide authentication using SHA-256 hashing. Pollard Rho factorization demonstrates vulnerability of small prime numbers. Security depends on difficulty of factoring large integers.